# Michael Wakeham

mwakeham@bu.edu
mikewakeham.github.io

## Research Interests

novel view synthesis, visual generative models, vision language models, controllable generation, world models, spatiotemporal scene understanding, 3D/4D reconstruction

## Education

**Boston University** — Boston, MA
*B.A. in Computer Science* — 2023 - 2027

- **Honors Thesis** (in progress): Spatial Intelligence in Vision-Language Models from Egocentric Infant Video
- **Graduate Coursework:** Deep Visual Generative Models, Advanced Topics in Computer Vision, Multimodal Machine Learning, Computer Vision, Deep Learning, Machine Learning, Computer Graphics

## Publications

1. **M Wakeham**, MK Das, W Wang, A Tewari, B Gong, D Ghadiyaram. "A 3D-Driven Framework for Moving-Camera Perspective-to-360° Video Generation". *IEEE/CVF Winter Conference on Applications of Computer Vision* (**WACV**), 2027
2. S Wang*, W Wang*, Z Wang*, M Whitton*, **M Wakeham**, A Chandra, J Huang, P Zhu, H Chen, D Li, JL Li, S Li, A Zagula, A Zhao, A Zhu, S Nakamura, Y Yamamoto, JJ Yokono, A Mueller, BA Plummer, K Saenko, V Saligrama, B Gong. "BabyVLM-V2: Toward Developmentally Grounded Pretraining and Benchmarking of Vision Foundation Models". *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR**), 2026

## Experience

**Research Assistant** | Boston University — May 2025 - present
*Advisors: Boqing Gong, Deepti Ghadiyaram*

- Proposed and developed a framework for panoramic video generation from perspective video via a 3D-driven formulation; published as first-author [1]
- Designed and built a cognitively-inspired benchmark for assessing spatiotemporal reasoning in VLMs via object motion in video clips; published as co-author [2]
- Received UROP Student Research Award for undergraduate research funding

**Visiting Research Intern** | Kempner Institute — July 2026 - present
*Advisors: Qianqian Wang, Ruojin Cai*

- Studying how to create a visuotactile object representation from pretrained visual shape priors (SAM3D) for dexterous manipulation tasks

## Teaching & Mentoring

**BU RISE Research Mentor** | Boston University — Summer 2026

- Mentored four high-school students conducting computer vision research on in-the-wild tracking of fish; resulted in NeurIPS 2026 workshop paper [link]

**Course Staff** | Boston University

- CS 541: Applied Machine Learning (Course Assistant) — Spring 2026
- CS 542: Machine Learning Principles (Grader) — Fall 2025